

Food and Agriculture Organization of United Nations

GeoNetwork OpenSource

a Standards based Geographic Data and Information Management System for the web

GeoNetwork's short history

- Prototyping by FAO (2000-2001)
- Version 1 by FAO & WFP (released 2003)
- Version 2 by FAO, WFP & UNEP (released 2005)
- Version 2.1 by FAO & UNOCHA (Oct 2007)
- Version 3.0 in planning phase (2008)

FAO

WFP

UNEP

UNOCHA

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GeoNetwork OpenSource: Geographic data sharing for everyone

Published by Allison Fleming

Modified over 5 years ago

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GeoNetwork OpenSource: Geographic data sharing for everyone
Welcome to FAO/NRCE
GeoNetwork is a Geographic Information Management System that enables easy and timely access to spatial information and cartographic products through descriptive metadata from a variety of data providers using the capacities of the Internet.
- 2

GeoNetwork Main Goals
Provide a common platform and standards to manage geographic data
Improve accessibility and sharing of a wide variety of geographic data at different scale and from multidisciplinary sources
Increase collaboration between UN and other international organizations for collecting data and make them available to the internet community
The main objectives of the project are:
to provide a common platform and standards to manage geographic data
to improve accessibility to data and information
to increase collaboration within and between organization for reducing duplication and enhancing information consistency and quality and data preparedness.
- 3

GeoNetwork's short history
Prototyping by FAO ()
Version 1 by FAO & WFP (released 2003)
Version 2 by FAO, WFP & UNEP (released 2005)
[Version 2.1 by FAO & UNOCHA \(Oct 2007\)](#)
Version 3.0 in planning phase (2008)
The prototype catalog system was developed in FAO in 2001 to systematically archive and publish the geographic datasets produced within the Organization.
Then WFP joined the project and with its contribution the first version of the software was released in 2003.
At that moment it was decided to develop the software as a Free and Open Source Software to allow the whole geospatial users community to benefit from the development results.
With UN Environmental Program (UNEP) a second version was released in 2005.
Now we are close to the latest version with the contribution of FAO and OCHA and the active support of CSI-CGIAR and UNEP
The version 3 is planned for next year
- 4

OGC Portal Reference Architecture
GeoNetwork has been developed based on the principles of a Free and Open Source Software and International Standards for services and protocols, like the ISO-TC211 and the OGC specifications. Its architecture is largely compatible with the OGC Portal Reference Architecture which is the OGC guide for implementing a standardized geospatial portal.
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Infrastructure Web Interface
GEONETWORK is an Internet Spatial Data Platform structured in three main modules
Web Interface
Metadata
Catalogue
The software's structure in fact relies on three main modules focusing on the GIS community needs:
the Metadata-Module
Spatial data Repository
Dynamic Map-Viewer
All of them accessible through a web interface provides
- an easy online metadata editor
- standard profiles for describing geodata
- data access through direct link to the data location
the Data-Module which is a Spatial data Repository facilitates
- timely data storage and exchange
- organization and management of geodata
the Dynamic Mapping-Intermap enables
- data visualization
- map production through query of multiple sources
Spatial Data Repository
- 6

GeoNetwork Main Functions
Metadata and data publication and distribution
Metadata and data search capabilities in distributed environment
Interactive access to maps
Administrative functions
Metadata editing and management
Different metadata standards
Different sharing levels
This slide summarizes the main functionalities of the software:
first of all the metadata and data publication and distribution, then the metadata search capabilities in distributed environment (I will explain later why the search can be distributed)
Another function is the visualization of the data on line,
the Administration functions to manage metadata and users,
the metadata editing and management using different standards and different levels of sharing
- 7

The Basic Building Blocks
ISO TC211 Standards
OGC Standards (Open Geospatial Consortium)
We said the software is based on Standards:
Standards as support for describing data and standards for searching and retrieving information from catalogs.
STANDARDS for VISUALIZING DATA. The web viewer integrated in GEONETWORK (INTERMAP) in fact is compliant with the OGC WMS and ArcIMS map services standards.
- 8

Metadata Standards Support for multiple metadata standards
ISO19115:2003
FGDC
Dublin Core
GeoNetwork supports the most common standards to specifically describe geographic data (ISO and FGDC) and the international standard for general documents (Dublin Core)
- 9

Standard Protocol for Catalog
Full support of the OGC Catalog Services for the Web (CSW)
It also fully compliant with the OGC specifications in the way it queries and retrieves information to and out of a web catalog
- 10

Metadata Harvesting and Synchronization
I would like to emphasize the role of the metadata harvesting within the concept of data sharing.
Through this functionality it is possible to collect information from the different GNW nodes installed around the world and to copy and store periodically this information locally. In this way a user from a single entry point can get information also from distributed databases.
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GeoNetwork Installations around the World
UN Organizations:
FAO, WFP, UNEP, WHO, OCHA, UNHCR
Other:
CGIAR, ICIMOD, ESA, GMFS, JRC, FGDC
Individual projects in Spain, France, Italy, Czech Republic, Australia, South Africa,
THIS IS THE web interface of the GEONETWORK OS SOFTWARE
and some examples of customization of the site in FAO, WFP and OCHA in NY.
- 12

The Home Page
It also fully compliant with the OGC specifications in the way it queries and retrieves information to and out of a web catalog
- 13

The Default Search
It also fully compliant with the OGC specifications in the way it queries and retrieves information to and out of a web catalog
- 14

The Search Results
It also fully compliant with the OGC specifications in the way it queries and retrieves information to and out of a web catalog
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Display Metadata
It also fully compliant with the OGC specifications in the way it queries and retrieves information to and out of a web catalog
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Display Metadata
It also fully compliant with the OGC specifications in the way it queries and retrieves information to and out of a web catalog
- 17

Display Interactive Map
Display Metadata
It also fully compliant with the OGC specifications in the way it queries and retrieves information to and out of a web catalog
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Open Map Viewer
It also fully compliant with the OGC specifications in the way it queries and retrieves information to and out of a web catalog
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Adding a Layer
It also fully compliant with the OGC specifications in the way it queries and retrieves information to and out of a web catalog
- 20

Exporting a Map
It also fully compliant with the OGC specifications in the way it queries and retrieves information to and out of a web catalog
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Map Layout
It also fully compliant with the OGC specifications in the way it queries and retrieves information to and out of a web catalog
- 22

The Advanced Search
It also fully compliant with the OGC specifications in the way it queries and retrieves information to and out of a web catalog
- 23

Food and Agriculture Organization of United Nations
Thanks
Patrizia Monteduro
Food and Agriculture Organization of United Nations
Rome, 27 September 2007

Similar presentations

On and Partners Metadata Training Program: 2008 GCP Project

Geospatial Metadata: Intermediate Course
Module 3: Metadata Catalog and Geospatial One Stop (GOS)

FAO

WFP

UNEP

UNOCHA

fgdc

Innovate!

2008

GEOS

EUROPEAN GEOGRAPHIC DATA SHARING ACTION PLAN

GEOS

EUROPEAN GEOGRAPHIC DATA SHARING ACTION PLAN

GEOSS

EuroGEOSS

A EUROPEAN APPROACH TO GEOS

EuroGEOS

Implementing the GEOS Data Sharing Action Plan: Drought

GEOS

EuroGEOS

Geospatial One-Stop

A Federal Gateway to
Federal, State & Local Geographic Data
<http://geodata.gov>

Spatial Data Infrastructure:
Concepts and Components

Geog 458: Map Sources and Errors
March 6, 2006

fgdc

Harvesting Metadata for Use
by the geodata.gov Portal

Doug Nebert
FGDC Secretariat
Geospatial One-Stop Team

INSPIRE

OneGeology-Europe - the first step to the
European Geological SDI

INSPIRE Conference 2010,
Session Thematic Communities: Geology
Krakow, June 24th 2010

John L. Landon
Agnes Tóthfalusi
Robert Tóthfalusi
Agnes Tóthfalusi
(1) - British Geological Survey (BGS)
(2) - French Geological Survey (BRGM)
(3) - Czech Geological Survey (CGS)

Introduction to Geospatial
Metadata – ISO 191** Metadata

National Coastal Data
Development Center
A Division of the
National Oceanographic Data Center

Please email a list of geospatial data
resources to:
metadatatool@noaa.gov
Also Email questions for the Q&A
session to metadatatool@noaa.gov

Planned Title: Review of
Evaluation of Geospatial Search

Allan Doyle

GeoNetwork

OpenSource

Spatial Data Management

GeoNetwork OpenSource:
Geographic data sharing for everyone

7 March 2008

FGDC Secretariat - GCP - Technical

AgriDrupal

- a "suite of solutions" for agricultural information
management and dissemination, built on the
Drupal CMS;
- the community of practice around these solutions.

Technical Review (2008, 2007)

data.gov 2.0 and Geospatial
Platform Integration

Doug Nebert
Senior Advisor for Geospatial Technology
CSI, FGDC Secretariat

Web-based Portal for
Discovery, Retrieval and Visualization
of Earth Science Datasets
in Grid Environment

Zhenping (Jane) Liu

GEOS

EuroGEOSS

A EUROPEAN APPROACH TO GEOS

EuroGEOS

Implementing the GEOS Data Sharing Action
Plan: Forestry

GEOS

EuroGEOS

CountrySTAT

The Global Administrative Unit Layers (GAUL)
Basic Concepts

Regional Training on Statistical Methods and Information Systems
Techniques for Improving CountrySTAT in the ECOT Region
Tehran, Iran 12 – 13 November 2014

Pablo Soto
CountrySTAT/FAO

Similar presentations

ISO

Geographic Information/Geomatics
Implementing
ISO Metadata

David Danko
Work Item 15 – Project Leader
ddanko@esri.com

Web Portal Presentation

None you to be added.

OGC SURVEY GUIDE
How To Build A Survey Portal

NTES

INSPIRE

The Geospatial Catalogue and Database Repository (CCDR)
and the Knowledge Management System (KMS)

The Geospatial Catalogue and Database Repository (CCDR)
and the Knowledge Management System (KMS)

Dr. Michael J. ...
Technical Officer - Information Systems and Data Management

GeoNetwork

ASIACOVER
Spatial Database Engine

Based on GeoNetwork
OpenSource Technology

ASIACOVER

GeoConnections

The GeoConnections Discovery Portal
Blair McLeod, Michael Adler
Patricia Monteduro, Michael Adler
MacDonald, Catherine and Associates

Geospatial One-Stop
FGDC and GOS
Working as One to Build the NSDI

Rob Dijkman
Geospatial One-Stop
Program Manager

countryData

CountryData Development
integrating the collection, update and dissemination of development
indicators (including the NSDI)

Wanda L. ...
2007

DFID

Geog 460
GIS Workshop

ArcGIS for Server

material assembled from the web pages of:
<http://www.esri.com/learn/arcgis/arcgis-server.pdf>
<http://www.esri.com/learn/arcgis/arcgis-server/arcgis-server/functional/matrix.pdf>

Tools for Creating and
Editing ISO Metadata

Various tools created depending on
user needs:
Desktop vs. Web Applications
GUI vs. XML

Design central EMODnet portal
Objectives, Technical Proposal and
Consultation Process

EMODnet

FGDC and GOS Metadata:
Foundations to Build the NSDI

Sharon ...
FGDC Secretariat / Geospatial One-Stop

ESIP & Geospatial One-Stop (GOS)

Registering ESIP Products and Services with
Geospatial One-Stop

GIS data sources:
catalogs of data and services

Geospatial One-Stop
FGDC and GOS
Working as One to Build the NSDI

Sharon ...
Federal Geographic Data Committee
Geospatial One-Stop
Metadata Coordinator

Registering Earth Science Data
and Data Related Services Using
NASA's Global Change Master
Directory (GCMD)

Tyler ...
USDA Forest Service
Technical Director, Forest SA
January 2, 2003

Distributed Data Analysis &
Dissemination System (D-DADS)

Special Interest Group
on Data Integration

June 2000

Prise en main de GeoNetwork:
gestion et publication de métadonnées
& harvesting et échange de
métadonnées Andrea de Bono
Certificat Géomatique - 16.02.2012.

Introducing ArcCatalog:
Tools for Metadata and
Data Management

Please Turn Off Your
Cell Phone and Pagers

ESIP Education User Conference - July 6 & 7, 2001

National Geospatial Enterprise
Architecture NSDI National Spatial
Data Infrastructure An Architectural
Process Overview Presented by Eliot
Christian.

May 2010 GGIM, New York City The
National System for Coordination of
Territorial Information NSDI of
Chile.

SDI for water resource management
Theresia Freska & Wu Liqun.

The Earth Information Exchange.
Portal Structure Portal
Functions/Capabilities Portal Content
ESIP Portal and Geospatial One-Stop
ESIP Portal and NOAA.

GCi Architecture

GEOS Information System Meeting
29 September 2013, ESA/ESRIN (Frascati, Italy)

N. ...
ESA/ESRIN

Grid Services for Digital Archive

Tao-Sheng Chen
Academia Sinica Computing Centre
tao-sheng.chen@sinica.edu.tw